MLR780 Air Conditioning and Refrigeration 2021

Assignment 3
Heating and Cooling Load Calculations

Instructions:

- This individual assignment is to be submitted online on or before **9 May 2021**
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment3-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment3-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcomes:

This assignment has been designed to assist you to understand the fundamentals of heating and cooling load calculations by means of the CLTD hand-based method of ASHRAE. This hand-based method is no longer the industry standard and has been replaced by more accurate computer-based methods. The CLTD method is, however, used in this module to demonstrate an overall method with which different cooling load aspects can be approximated.

Background:

Consider a portion of the Engineering 4.0 building on the Innovation Africa Campus of the University of Pretoria, shown schematically from above in Figure 1. You are required to do the cooling load calculations for Building A (shown in red). For simplicity, the approximate outline of the building indicated as a blue dashed line should be used.

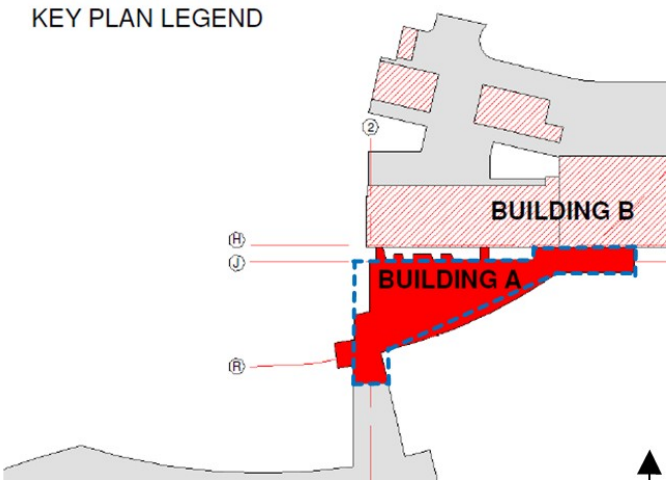


Figure 1 Relative placement of the building.

Before you start, visit this site: https://www.up.ac.za/eng4 and view the video from 02:16 for visuals of the building, which you can use to assist you in your assumptions.

Several PDF files also accompany this assignment containing the actual dimensions and details of the building, which were used during construction. The most important ones to consult first are:

- Drawing 151 (floor plan)
- Drawing 231 (elevations)
- Drawing 200 (main sections)

  Note: these PDF's contain a VERY large amount of data and might take some time to open. If your PDF viewer is not able to render the content, open the files with an internet browser (such as Firefox etc.)

Once you are relatively familiar with the most important and RELEVANT information on these, you can consult the other documents if needed.

Some material specifications documents are also provided.

For purposes of this assignment, please centre your analysis around the following:

- Treat the entire building as a single zone (defined by the blue outline in Figure 1).
- External temperature is based on ASHRAE 2009 **1%** weather data.
- Wind speed is based on ASHRAE 2009 **5%** annual extreme wind speed.
- Indoor design condition: Summer and winter: **22ºC** dry bulb and **50%** relative humidity.
- The building is designed to be occupied on busy days from **07:30** to **18:30** and will host the following functions:
  - Offices, each with 1 occupant, 1 standard type desktop computer, 2 screens.
  - An auditorium (occupants may have laptops and smart devices)
  - A kitchen with a standard type fridge/freezer, a standard type microwave, and a standard type wall mounted hot water urn.
  - Walkways and study areas designed to house approximately 100 people with their laptops and smart devices comfortably (this excludes the occupants in the auditorium).
  - Several other smaller rooms (including ablutions etc), for which you can ignore any internal heat sources.
- Facades are a combination of glass, palram polycarbon sheeting (10mm) and precast concrete as indicated in the elevations.
- Glass:
  - South: Solar E Evergreen glass (single glazing at 6 mm) – See glazing specification page 2
  - West and North: Solar E Evergreen glass (double glazing at 6 mm + 12 mm air + 6 mm glass) – See glazing specification page 3
- Roof insulation: 50 mm
- Lights: LED lights with a total power consumption of 8 kW.

- The adjacent building has the same internal temperature.
- Assume a latitude of **40° South** (This is not the real latitude for Pretoria, but used for calculation purposed only)
- Assume a direct (**100%** outdoor air) ventilation rate equal to the minimum recommended fresh air rate as prescribed in the ASHREA ventilation standard.
- Assume infiltration to be negligible.
- Ignore any external shading devices, or shading from adjacent structures.
- Make an appropriate assumption if required values or data are not supplied (show and shortly motivate your assumptions).
- **If any additional information is made available, it will be listed on the MLR780 discussion board.**

Assignment / Calculations:
Use the CLTD cooling load procedure discussed in class to estimate the cooling load. For purposes of this assignment only use CLTD related tables as supplied in Chapter 28 of ASHRAE 1997.

Perform the Cooling Load Calculation (Sensible and Latent) for the zone for normal operating conditions. Structure your assignment as follows:

- Weather data
- Material properties for all relevant external windows, walls, doors, floors, and ceiling and overall heat transfer coefficients [W/m^2K] for each component type.
- Estimate the peak cooling hour
- To help the lecturer identify your marks, please work systematically and address the items given below (show method) in the listed sequence. If relevant, split each category into the relevant sub-calculations (for instance: North window**s**, east window**s** etc, or: Occupants office, occupants auditorium etc..). Use headings please.

Tabulate a summary of your final answers as follows:

	Value
Roof/ceiling	
Walls	
Windows	
Doors	
Floor	
Radiation through windows.	
Internal equipment heat sources	
Internal light heat sources	
Internal Occupants	
Ventilation and infiltration.	
Total Cooling Load	

INDICICATE HOW MUCH TIME YOU SPENT ON THIS ASSIGNMENT.